

Mohammad Rafiqul Islam

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ACADEMIC APPOINTMENT

Postdoctoral Research Associate Iowa State University, Ames, Iowa Postdoc Mentor: <i>Professor David Herzog</i>	August 2026 – To Date
Graduate Teaching Assistant Florida State University, Tallahassee, Florida	August 2021 – July 2026
Graduate Assistant Youngstown State University, Youngstown, Ohio	August 2018 – May 2020

EDUCATION

Ph.D. in Mathematics Florida State University, Tallahassee, Florida Dissertation: <i>Advances in Langevin Algorithms for Decentralized, Constrained, and Accelerated Sampling.</i> Advisor: <i>Professor Lingjiong Zhu</i>	July 2026
M.S. in Mathematics Youngstown State University, Youngstown, Ohio	May 2020
M.S. in Applied Mathematics University of Dhaka, Dhaka, Bangladesh	March 2016
B.S. in Mathematics University of Dhaka, Dhaka, Bangladesh	November 2014
Society of Actuaries (SOA) Exam P, Exam FM	January 2019

RESEARCH INTERESTS

Stochastic Optimization and Sampling: Markov chain Monte Carlo (MCMC) methods, Langevin algorithms, decentralized Bayesian inference, constrained sampling, and high-dimensional probability.

Financial Engineering: Predictive modeling techniques for time series data using classical statistical and machine learning models; derivative pricing; portfolio risk management.

PUBLICATIONS

PREPRINTS

- [1] **Islam, Mohammad Rafiqul**, and Lingjiong Zhu. “*Decentralized Proximal Stochastic Gradient Langevin Dynamics.*” arXiv preprint arXiv:2605.00723 (2026).
- [2] Dang, Thanh, Mert Gürbüzbalaban, **Mohammad Rafiqul Islam**, Nian Yao, and Lingjiong Zhu. “*High-Order Langevin Monte Carlo Algorithms.*” arXiv preprint arXiv:2508.17545 (2025).
- [3] Gürbüzbalaban, Mert, **Mohammad Rafiqul Islam**, Xiaoyu Wang, and Lingjiong Zhu. “*Generalized EXTRA stochastic gradient Langevin dynamics.*” arXiv preprint arXiv:2412.01993 (2024).

PEER-REVIEWED PUBLICATIONS

- [4] Mostafa, Fahad, Pritam Saha, **Mohammad Rafiqul Islam**, and Nguyet Nguyen. “*GJR-GARCH volatility modeling under NIG and ANN for predicting top cryptocurrencies.*” *Journal of Risk and Financial Management* 14, no. 9 (2021): 421.
- [5] **Islam, Mohammad Rafiqul**, and Nguyet Nguyen. “*Comparison of financial models for stock price prediction.*” *Journal of Risk and Financial Management* 13, no. 8 (2020): 181.

PRESENTATIONS AND TALKS

- [6] *Scalable Sampling Algorithms via Langevin Dynamics*; **Talk**; Analysis, Geometry and Probability Seminar, Department of Mathematics, Iowa State University; September 9, 2026.
- [7] *Decentralized Constrained Sampling via Proximal Stochastic Langevin Dynamics*; **Contributed talk**; AMS 2026 Spring Eastern Sectional Meeting, Boston College, MA; March 28 – 29, 2026.
- [8] *Generalized EXTRA SGLD: Eliminating Network Bias in Decentralized Bayesian Learning*; **Poster presentation**; Bayesian Inverse Problems and UQ, ICERM, Brown University, Providence, RI; March 2 – 6, 2026.
- [9] *High-Order and Bias-Free Distributed Langevin Dynamics for Scalable Bayesian Inference*; **Talk**; Department of Mathematics, Florida State University; January 30, 2026.
- [10] *The Heavy-Tail Phenomenon in Decentralized Stochastic Gradient Descent*; **Talk**; Department of Mathematics, Florida State University; November 20, 2023.
- [11] *Decentralized Stochastic Gradient Langevin Dynamics and Hamiltonian Monte Carlo*; **Talk**; Department of Mathematics, Florida State University; October 5, 2023.
- [12] *Asymptotics and calibration of local volatility models*; **Talk**; Department of Mathematics, Florida State University; December 2, 2022.
- [13] *Sensitivity analysis for Monte Carlo and Quasi Monte Carlo option pricing*; **Talk**; Department of Mathematics, Youngstown State University; April 28, 2020.

TEACHING EXPERIENCE

Department of Mathematics August 2021 – July 2026
Florida State University, Tallahassee, Florida.

Instructor of Record

1. MAP 2302 Ordinary Differential Equations; Summer 2026.
2. MAC 2312 Calculus and Analytic Geometry II; Fall 2025, Spring 2026.
3. MAC 2311 Calculus and Analytic Geometry I; Spring 2025, Summer 2025.
4. MAP 4170 Introduction to Actuarial Mathematics (Financial Mathematics for Actuaries); Spring 2024.
5. MAC 1140 Precalculus Algebra; Spring 2023.

Recitation Instructor and Grader

6. MAC 2311 Calculus and Analytic Geometry I; Fall 2022.
7. MAP 4170 Introduction to Actuarial Mathematics (Financial Mathematics for Actuaries); Fall 2024, Fall 2025.

Department of Mathematics and Statistics August 2018 – May 2020
Youngstown State University, Youngstown, Ohio.

Instructor of Record

8. MATH 1511 Trigonometry; Fall 2019, Spring 2020.
9. MATH 1510 College Algebra; Fall 2018.

UNDERGRADUATE RESEARCH MENTORING

Research Mentor Fall 2025 – Spring 2026

FSU Undergraduate Research Opportunity Program (UROP)

- Mentored Sarah Thapa on a predictive-modeling project examining factors associated with Florida road crashes and the economic impact of crash-related casualties.
- Mentored Aiden Marin on an option-pricing project comparing the performance of the classical Black–Scholes–Merton model and modern machine-learning models for pricing options on different underlying stocks.

Research Mentor Fall 2024 – Spring 2025

FSU Mathematics Directed Reading Program (DRP)

Mentored two undergraduate students, *Lily Kennedy* and *Kiera Agic*, in the study and implementation of machine-learning methods for an NLP project, “Identifying Spam and Non-Spam Emails.”

HONORS AND AWARDS

1. Tam Family Excellence in Graduate Student Research Award, Florida State University. April 2026.

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| 2. Distinguished Teaching Assistant Award, Florida State University. | April 2026. |
| 3. Tam Family Poster Competition Award, Second Place, Florida State University. | April 2026. |
| 4. Bettye Anne Busbee Case Graduate Fellowship and Doctoral Mentorship Recognition. | April 2024. |
| 5. Outstanding Graduate Student in Statistics Award, Youngstown State University. | April 2020. |
| 6. Graduate College Premiere Scholarship, Youngstown State University. | August 2018. |
| 7. MetLife Bangladesh Actuarial Study Program Fellowship. | January 2015. |

SELECTED RESEARCH AND TRAVEL FUNDING

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| 8. American Mathematical Society (AMS) Graduate Student Travel Grant. | March 2026. |
| 9. Institute for Computational and Experimental Research in Mathematics (ICERM) Workshop Grant. | March 2026. |
| 10. Florida State Undergraduate Research Opportunity Program Mentorship Fund. | March 2026. |
| 11. Florida State University Mathematics Department Travel Grant. | March 2026. |
| 12. Congress of Graduate Students Conference Presentation and Attendance Grant. | March 2026. |

INDUSTRY EXPERIENCE

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| Assistant Vice President
Delta Life Insurance Company Ltd., Dhaka, Bangladesh | Sept. 2017 – July 2018 |
| Actuarial Trainee Executive
Sandhani Life Insurance Company Ltd., Dhaka, Bangladesh | Sept. 2016 – August 2017 |

CONFERENCES, WORKSHOPS, AND SEMINARS

- [1] American Mathematical Society (AMS) 2026 Spring Eastern Sectional Meeting (March 28 – 29, 2026), Boston College, Massachusetts.
- [2] Bayesian Inverse Problems and Uncertainty Quantification (March 2 – 6, 2026), The Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, Rhode Island.
- [3] 1st SRU SAS Analytics Day, Slippery Rock University and SAS Institute, March 2019.
- [4] 6th Annual Midwest Actuarial Student Conference, Society of Actuaries and the University of Wisconsin, Milwaukee, Wisconsin, November 2018.
- [5] 20th International Mathematics Conference, Bangladesh Mathematical Society, Dhaka, Bangladesh, 2017.

MEMBERSHIPS

- Bangladesh Mathematical Society: Life Member
- Society of Actuaries (SOA), USA: Student Member
- Society for Industrial and Applied Mathematics (SIAM), USA.
- American Mathematical Society (AMS), USA.

REFERENCES

Lingjiong Zhu, Ph.D.

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